31.1 Faraday's Law of Induction :-

• Induced EMF & Current :-

A changing magnetic field in a circuit induces an emf. The current that flows due to this EMF is called an induced current. This happens even without battery.

• Key Principle :-

An EMF is induced only when the magnetic flux through a loop changes with time. A constant, steady magnetic flux does not induce an EMF.

• Faraday's Law of Induction :-

"The electromotive force (EMF) induced in a closed-loop is directly proportional to the rate of change of magnetic flux (Φ_B) passing through the loop".

• Mathematically : $\mathcal{E} = -\frac{d\Phi_B}{dt}$

where $\Phi_B = \oint \vec{B} \cdot d\vec{A}$ is the magnetic flux through the loop. The negative sign significance is given by Lenz's law.

- Formula for a coil of N loops:

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}$$

• Ways to Induce an EMF :-

Since the magnetic flux is $\Phi_B = BA\cos(\theta)$, an EMF can be induced by changing any of these factors with time.

1. The magnitude of the magnetic field (B).
2. The area (A) of the loop.
3. The angle between the magnetic field B and the normal to the loop (θ).

31.3 Lenz's Law :-

This law answers: "Which direction does the induced current flow?".

- Statement :-

"The induced current in a loop will always flow in a direction that creates its own magnetic field ($B_{induced}$) to oppose the change in magnetic flux that produced it."

- Why ? Conservation of energy. The universe resists change. If the induced current helped the change, it would create a runnway feedback loop and free energy, which is impossible.

- **Key Point to Remember :-**

- If flux is increasing : The induced current creates a magnetic field to oppose the increase. (It creates an opposite field).

- If flux is decreasing : The induced current creates a magnetic field to support the decrease. (It creates a same field).

* 31.2 - Motional EMF :-

- **Definition :-**

Motional EMF ($\mathcal{E}$) is induced in a conductor of length l when it moves with a velocity v through a constant magnetic field B.

$$\mathcal{E} = B . l . v$$

$\mathcal{E}$ = Motional EMF
B = magnetic field strength
l = length of the conductor
v = velocity of the conductor.

$$\Delta V = \frac{W}{q} = \frac{F.l}{q} = \frac{(Eq)l}{q} = El \Rightarrow \Delta V = El$$

1. Derivation (Straight Conductor):-

- Consider a conductor of length l moving with velocity v through a perpendicular magnetic field.
- The positive and negative charges q inside the conductor are also moving at velocity v.
- These charges feel a magnetic force (Lorentz Force): $F_B = qv \times B$. This force pushes electrons to one end of the conductor.
- Charges separation creates an internal electric field E.
- Charges stop moving when the new downward electric force ($F_E = qE$) balances the upward magnetic force ($F_B = qvB$).

$$\text{Electric Force} = \text{Magnetic Force}$$
$$qE = qv \times B$$
$$E = vB$$

The potential difference (voltage) across the ends is the motional EMF $\mathcal{E} = \Delta V = El$.

$$\boxed{\mathcal{E} = Blv}$$

2. Derivation (Sliding Bar on Rails) :-

- Consider a sliding bar (length l) on rails. It forms a loop with area $A = lx$, where x is the horizontal position.

- Magnetic flux : $\Phi_B = BA = Blx$

- Using Faraday's Law : $\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}(Blx)$

- B and l are constant, so : $\mathcal{E} = -Bl\left(\frac{dx}{dt}\right)$.

- We know that $v = \frac{dx}{dt}$, so we replace dx/dt with v.

$$\boxed{\mathcal{E} = -Blv}$$

The induced current in the circuit (with resistance R) is :

$$\boxed{I = \frac{|\mathcal{E}|}{R} = \frac{Blv}{R}}$$

3. Derivation (The Rotating Bar) :- Numericals

- Consider a bar of length (l) rotates with constant angular speed (ω) in a magnetic field (B).

$v = r\omega$

- Consider a tiny piece of the bar dr at a distance r from the pivot.
- It's speed is not v, but $v = r\omega$.
- Now we can use our Motional EMF formula ($\mathcal{E} = Blv$), but only for this tiny piece dr.

$$d\mathcal{E} = Bv\,dr$$

- Substitute v: $d\mathcal{E} = B(r\omega)\,dr$
- To find the total emf, we sum up (integrate) all the tiny emfs from the pivot ($r = 0$) to the end ($r = l$).

$$\mathcal{E} = \int d\mathcal{E} = \int_0^l B\omega r\,dr$$

- B and ω are constants; so pull them out.

$$\mathcal{E} = B\omega \int_0^l r\,dr$$

$$\mathcal{E} = B\omega \left[\frac{1}{2}r^2\right]_0^l = B\omega\left(\frac{1}{2}l^2 - 0\right)$$

$$\boxed{\mathcal{E} = \frac{1}{2}B\omega l^2}$$

31.4 – Induced EMF & Electric Fields

- A changing magnetic field or flux creates an induced electric field.

- This induced electric field is nonconservative, meaning the work it does on a charge over a closed path is not zero.

• General Form of Faraday's Law :-

This equation relates the nonconservative electric field to the changing magnetic flux.

$$\oint E \cdot ds = -\frac{d\Phi_B}{dt}$$

• Derivation : E-Field of a Solenoid :-

Problem : A long solenoid with radius R has a B-field that is changing at a rate $\frac{dB}{dt}$. Find the induced E-field.

- By symmetry, the induced E-field lines are circles.
- The left side of the law is $\oint E \cdot ds = E(2\pi r)$.
- Case 1 : Inside the solenoid ($r < R$) :
- We use an imaginary loop of radius r.

- The flux inside this path is $\Phi_B = B \cdot A = B(\pi r^2)$

$$\oint E \cdot ds = -\frac{d\Phi_B}{dt}$$

$$E(2\pi r) = -\frac{d}{dt}(B\pi r^2)$$

$$E(2\pi r) = -(\pi r^2)\frac{dB}{dt}$$

$$E = -\frac{r}{2}\frac{dB}{dt}$$

The E-field grows linearly from the center.

- Case 2: Outside the solenoid ($r > R$):

- We use an imaginary loop of radius $r > R$.

- The flux inside this path is only where the B-field exists (i.e inside R). So, $\Phi_B = B \cdot A_{solenoid} = B(\pi R^2)$.

$$\oint E \cdot ds = -\frac{d\Phi_B}{dt}$$

$$E(2\pi r) = -\frac{d}{dt}(B\pi R^2)$$

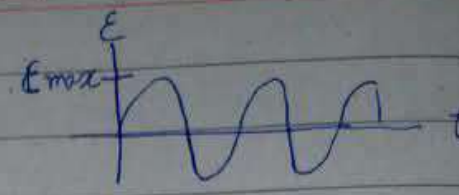
$$E(2\pi r) = -(\pi R^2)\frac{dB}{dt}$$

$$E = -\frac{R^2}{2r}\frac{dB}{dt}$$

The E-field gets weaker as $1/r$.

31.5 – Generators And Motors

• AC Generators :-



- Function : Converts Mechanical Energy into Electrical Energy.

- How : A coil is mechanically rotated (with angular speed ω) in a constant B-field, causing the flux to change by changing the angle θ and inducing an EMF.

• Derivation : The emf of a Generator :-

- A coil with N turns and area A rotates at angular speed ω in a field B. The angle at time t is $\theta = \omega t$

- Flux : $\Phi_B = BA\cos\theta = BA\cos(\omega t)$

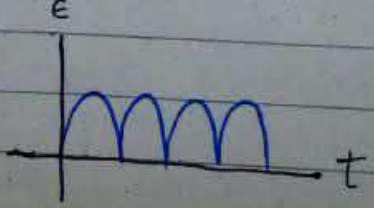
- We have set the clock so that $t = 0$ when $\theta = 0$. Hence the induced emf in the coil is.

$$\mathcal{E} = -N\frac{d\Phi_B}{dt} = -N\frac{d}{dt}(BA\cos(\omega t)) = -NAB(-\omega\sin\omega t)$$

$$\mathcal{E}(t) = NAB\omega\sin(\omega t)$$

Maximum EMF ($\mathcal{E}_{max}$) occurs when $\sin(\omega t) = 1$

$$\mathcal{E}_{max} = NAB\omega$$



• DC Generator :-

Similar to an AC generator but uses a split ring commutator to reverse the connection every half-turn, ensuring the output voltage always has the same polarity.

• Motors :-

• Function: Converts Electrical Energy into Mechanical Energy. Instead of generating a current by rotating a coil, a current is supplied to the coil by a battery and the torque acting on the current-carrying ~~conductor~~ coil causes it to rotate.

• As the motor's coil spins (ω), it also acts as a generator.
• This spinning creates its own induced emf, $\mathcal{E}_{back}$.
• By Lenz's Law, this back emf opposes the applied voltage $\mathcal{E}_{applied}$.
• The net voltage that actually pushes current through the motor is $\Delta V_{net} = \mathcal{E}_{applied} - \mathcal{E}_{back}$

• The current drawn by the motor is

$$I = \frac{\Delta V_{net}}{R} = \frac{\mathcal{E}_{applied} - \mathcal{E}_{back}}{R}$$

* At startup : $I = \mathcal{E}_{applied} / R$.
* At Jammed : $I = \mathcal{E}_{applied} / R$.

31.6 – Eddy Currents

Definition :-

These are simply induced currents (following Lenz's Law) that flow in bulk pieces of metal (like a sheet or block) instead of a thin wire. When a metal sheet moves through a B-field, the flux through it changes. This induces swirling loops of current (eddy currents) within the metal.

- The Effect: These currents create their own B-fields that, by Lenz's Law, always oppose the motion. This creates a "magnetic drag" or braking force.
- Applications: Electromagnetic braking on trains and roller coasters (smooth, frictionless stopping).
- Reduction: To reduce unwanted loss of energy from eddy currents, conducting parts are often laminated (built from thin layers separated by an insulator) to break up the large current loops.

Ch # 32 : Inductance

32.1 – Self Inductance & Inductance

• Concept of Self - Induction :-

- When current flows through a coil, it creates a magnetic field, and therefore a magnetic flux through the coil itself.
- If the current changes with time, the flux also changes.
- By Faraday's Law, this changing flux induces an emf $\mathcal{E}_L$ in the same coil this phenomenon is called self - induction.
- The induced emf $\mathcal{E}_L$ is called a self - induced emf or back emf.

• Inductance :-

Inductance is the property of a circuit element (like a coil) that describes how much back emf is induced for a given rate of change of current. It's a measure of the element's opposition to a change in current.

"The self-induced emf $\mathcal{E}_L$ is always proportional to the rate of change of the current $\frac{dI}{dt}$. The constant of proportionality is L, the inductance."

$$\mathcal{E}_L = -L \frac{dI}{dt}$$

- **Formula From Geometry** :- Inductance can also be defined based on the physical properties of the coil.

$$L = \frac{N \Phi_B}{I}$$

- A circuit element designed to have specific inductance is called an inductor. Its main purpose to be "sluggish" and resist any changes in the current.

Unit : The SI unit of ~~resistance~~ inductance is Henry (H).

1 H = 1 Volt . second / Ampere.

★ Resistance (R) is the opposition to current (I).
★ Inductance (L) is the opposition to change in current $\left(\frac{dI}{dt}\right)$.

32.2 – RL Circuits

- **RL Circuit**: A circuit containing a resistor (R) and an inductor (L).

- **Inductor's Role**: An inductor opposes any change in current.

- When you try to start a current, the inductor creates a back emf to fight the increase, making the current rise gradually.

- When you try to stop a current, the inductor creates an emf to fight the decrease, making the current fall gradually.

- **Time Constant (τ)**: The key parameter for an RL circuit. It defines the time (in seconds) it takes for the current to reach 63.2% ($1 - e^{-1}$) of its final value. It is defined as:

$$\tau = \frac{L}{R}$$

- **Derivation: Current in a Charging RL Circuit**:

Consider a series circuit with a resistor R and inductor L, connected with a battery of emf $\mathcal{E}$. When the switch is closed at $t = 0$, the current begins to increase, but the inductor's back emf opposes the rise.

We can describe the circuit by applying Krichoff's Loop Rule.

$$\mathcal{E} - \Delta V_R - \Delta V_L = 0$$

Substituting the expression for the voltage drops across the resistor (IR) and the inductor ($L\frac{dI}{dt}$); we get the differential equation for the current.

$$\mathcal{E} - IR - L\frac{dI}{dt} = 0$$

To solve for the current $I(t)$, we must separate the variable I and t.

$$\mathcal{E} - IR = L\frac{dI}{dt}$$

$$\frac{dt}{L} = \frac{dI}{\mathcal{E} - IR}$$

Next, we integrate this expression. At time $t = 0$, the current is 0. At some later time t the current is I.

$$\int_0^I \frac{dI}{\mathcal{E} - IR} = \int_0^t \frac{dt}{L}$$

$$-\frac{1}{R}\int_0^I \frac{dI}{\mathcal{E} - IR} = \frac{t}{L}$$

$$\left[-\frac{1}{R} \ln(\mathcal{E} - IR) \right]_0^I = \frac{t}{L}$$

$$-\frac{1}{R} \left[\ln(\mathcal{E} - IR) - \ln(\mathcal{E} - 0) \right] = \frac{t}{L}$$

$$\ln\left(\frac{\mathcal{E} - IR}{\mathcal{E}} \right) = -\frac{Rt}{L}$$

To isolate I, we take exponential of both sides.

$$\frac{\mathcal{E} - IR}{\mathcal{E}} = e^{-\frac{R}{L}t}$$

$$1 - \frac{IR}{\mathcal{E}} = e^{-\frac{R}{L}t}$$

Substitute $T = L/R$

$$\frac{IR}{\mathcal{E}} = 1 - e^{-\frac{t}{T}}$$

This gives the final expression for current (I) as a function of time (t).

$$\boxed{I(t) = \frac{\mathcal{E}}{R}\left(1 - e^{-t/T}\right)}$$

Current in a Discharging RL Circuit :-

This derivation finds the current $I(t)$ in an RL circuit after the battery has been removed. We assume the circuit was fully charged and had an initial current $I_i = \mathcal{E}/R$ at $t = 0$

We can describe the new circuit by applying Kirchoff's loop rule. The ↓ emf $\mathcal{E}$ is now 0 :
battery

$$0 - IR - L\frac{dI}{dt} = 0$$

To solve for current $I(t)$, we must separate the variables I and t.

$$L\frac{dI}{dt} = -IR$$

$$\frac{dI}{I} = -\frac{R}{L}\,dt$$

Next, we integrate this expression. At time $t = 0$, the current is at its maximum initial value I_i. At some later time t, the current is I.

$$\int_{I_i}^{I} \frac{dI}{I} = \int_0^t -\frac{R}{L}\,dt$$

$$\int_{I_i}^{I} \frac{1}{I}\,dI = -\frac{R}{L}\int_0^t dt$$

$$\left[\ln I\right]_{I_i}^{I} = -\frac{R}{L}\left[t\right]_0^t$$

$$\ln(I) - \ln(I_i) = -\frac{R}{L}(t - 0)$$

$$\ln\left(\frac{I}{I_i}\right) = -\frac{R}{L}t$$

To solve for I, we take the exponent of both sides.

$$\frac{I}{I_i} = e^{-\frac{R}{L}t}$$

$$I = I_i\, e^{-\frac{R}{L}t}$$

We substitute the time constant of the RL circuit $\tau = L/R$ and the initial current $I_i = \mathcal{E}/R$. This gives the final expression for the current I as the function of time t:

$$\boxed{I(t) = \frac{\mathcal{E}}{R}\, e^{-\frac{t}{\tau}}}$$

32.3 - Energy in a Magnetic Field.

An inductor stores energy in its magnetic field. This energy is supplied by the battery. To push current I into the inductor against its own back emf ($\mathcal{E}_L$) the battery must do work. This work is stored as potential energy in the inductor's magnetic field. The total energy U_L stored in an inductor L carrying current I is:

$$U_L = \frac{1}{2} L I^2$$

- Magnetic Energy Density (u_B): This is the energy stored per unit volume (in J/m^3). It is a property of the magnetic field itself.

$$u_B = \frac{B^2}{2\mu_0}$$

Derivation: Energy Stored in an Inductor :-

To find the energy stored in an inductor, we consider the energy balance in an RL circuit. Multiplying the loop rule equation by the current I gives us the power equation of the circuit.

$$I\mathcal{E} = I(IR) + I\left(L\frac{dI}{dt}\right)$$

$$I\mathcal{E} = I^2R + LI\frac{dI}{dt}$$

Each term represents a form of power.

- $I\mathcal{E}$ = Total power supplied by the battery.
- I^2R = Power dissipated as heat in the resistor.
- $LI\frac{dI}{dt}$ = The rate at which energy is being stored in the inductor.

Let U be the energy stored in the inductor. The rate of energy storage is $\frac{dU}{dt}$.

$$\frac{dU}{dt} = LI\frac{dI}{dt} \quad , \quad dU = LI\,dI$$

To find the total energy stored as the current increases from 0 to the final value I, we integrate:

$$\int_0^U dU = \int_0^I LI\,dI$$

$$U = L\left[\frac{1}{2}I^2\right]_0^I = \left(\frac{1}{2}LI^2 - 0\right)L$$

This give the ~~final~~ formula for the energy stored in the magnetic field of an inductor.

$$\boxed{U_L = \frac{1}{2}LI^2}$$

. Derivation: Magnetic Energy Density

Energy stored per unit volume is called energy density. We can derive this by using the example of a solenoid.

The energy stored in a solenoid is $U = \frac{1}{2}LI^2$

We know that the inductance of a solenoid is:

$$L = \mu_0 n^2 V$$

We also know the magnetic field of solenoid is:

$B = \mu_0 n I$, which means $I = \frac{B}{\mu_0 n}$

Substitute the value of L and I into the energy formula:

$$U = \frac{1}{2}(\mu_0 n^2 V)\left(\frac{B}{\mu_0 n}\right)^2$$

$$U = \frac{1}{2}(\mu_0 n^2 V)\left(\frac{B^2}{\mu_0^2 n^2}\right)$$

$$U = \frac{1}{2}\frac{B^2 V}{\mu_0}$$

The magnetic energy density, u_B, is the energy per unit volume, U/V:

$$u_B = \frac{U}{V} = \frac{B^2 V}{2\mu_0}\left(\frac{1}{V}\right)$$

$$u_B = \frac{B^2}{2\mu_0}$$

This gives the general formula for the energy density in any magnetic field:

$$\boxed{u_B = \frac{B^2}{2\mu_0}}$$

• Key Derivations :-

1) Emf in coil 2 (due to I_1):

Start with Faraday's law for coil 2:

$$\mathcal{E}_2 = -N_2 \frac{d\Phi_{12}}{dt}$$

From the definition, we know $N_2 \Phi_{12} = M I_1$

Substitute this into Faraday's law:

$$\mathcal{E}_2 = -\frac{d}{dt}(M I_1)$$

Since M is a constant:

$$\boxed{\mathcal{E}_2 = -M \frac{dI_1}{dt}}$$

2) Emf in coil 1 (due to I_2):

The process is identical.

$$\mathcal{E}_1 = -\frac{N_1 d\Phi_{21}}{dt}$$

We know $N_1 \Phi_{21} = M I_2$

$$\boxed{\mathcal{E}_1 = -M \frac{dI_2}{dt}}$$

32.4 - Mutual Inductance

- **Mutual Induction:** If two coil are placed together near each other. A changing current I_1 in one coil creates a changing magnetic field, which in turns creates a changing magnetic flux Φ_{12} through a second nearby coil. This changing flux induces an emf $\mathcal{E}_2$ in the second coil. This is the principle behind transformers and wireless charging.

- **Mutual Inductance (M):** A constant that measures how effectively a changing current in one coil induces an emf in another. It depends only on the geometry and relative position of the two coils.

. The mutual inductance of coil 2 with respect to coil 1 is:

$$M_{12} = \frac{N_2 \Phi_{12}}{I_1}$$

(Where Φ_{12} is the flux in coil 2 created by I_1 in coil 1.

. The mutual inductance of coil 1 with respect to coil 2 is:

$$M_{21} = \frac{N_1 \Phi_{21}}{I_2}$$

Unit is Henry (H).

$$0 = \frac{Q}{C}\frac{dQ}{dt} + LI\frac{dI}{dt}$$

- We know $I = \frac{dQ}{dt}$ and $\frac{dI}{dt} = \frac{d^2Q}{dt^2}$. Substitute these into the equations.

$$\frac{Q}{C}(I) + L(I)\frac{d^2Q}{dt^2} = 0$$

- Divide the whole equation by I (since I is not always zero).

$$\frac{Q}{C} + L\frac{d^2Q}{dt^2} = 0$$

$$\boxed{\frac{d^2Q}{dt^2} = -\frac{1}{LC}Q}$$

- This is the differential equation for an LC circuit.

- The equation $\frac{d^2Q}{dt^2} = -\left(\frac{1}{LC}\right)Q$ is identical in form to the equation for a simple harmonic oscillator, $\frac{d^2x}{dt^2} = -\omega^2 x$.

- By comparing the two equations, we get:

$$\boxed{\omega = \frac{1}{\sqrt{LC}}}$$

32.5 - Oscillations in an LC Circuit :-

- **LC Circuit:** An idealized circuit (assuming 0 resistance) containing only an inductor (L) and a capacitor (C).

- **Oscillations:** The circuit creates spontaneous electromagnetic oscillations, where energy is continuously transferred back and forth between the electric field of the capacitor and the magnetic field of the inductor.

- **Derivation: Differential Equation for An LC Circuit :-**

  - We use the Conservation of Energy. In an ideal LC circuit (no resistor), the total energy U is constant, so its rate of change $\frac{dU}{dt}$ is zero.

  - The total energy is the sum of energy in the capacitor and the inductor.

$$U = U_C + U_L = \frac{Q^2}{2C} + \frac{1}{2} L I^2$$

  - Now, we differentiate the energy equation with respect to time :

$$\frac{dU}{dt} = \frac{d}{dt}\left(\frac{Q^2}{2C} + \frac{1}{2} L I^2\right)$$

$I_{max} = Q_{max} \cdot \omega$ $\qquad f = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$.

$\omega = \frac{1}{\sqrt{LC}}$, $\omega = 2\pi f$

. The solution to this equation is a sinusoidal function. For a capacitor that starts at Q_{max} at $t = 0$, the solution is:

$$\boxed{Q(t) = Q_{max} \cos(\omega t)}$$

. For the current I as a function of time.

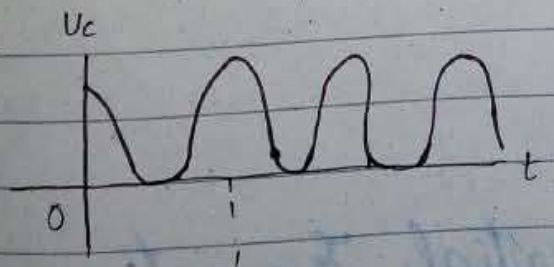
$$I(t) = \frac{dQ}{dt}$$

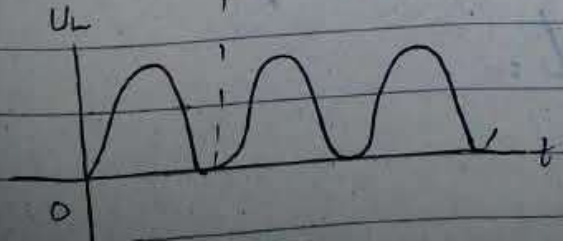
$$I(t) = \frac{d}{dt}\left(Q_{max} \cos(\omega t)\right)$$

$$I(t) = -\omega Q_{max} \sin(\omega t)$$

. We define the maximum current I_{max} as ωQ_{max}.

$$\boxed{I(t) = -I_{max} \sin(\omega t)}$$





- We start with the conservation of energy, but this time we account for the energy lost to the resistor.

- The total energy $U = U_C + U_L$ is not constant. The rate at which the total energy changes, $\frac{dU}{dt}$, must be equal to the rate at which energy is dissipated by the resistor, $-I^2R$.

$$\frac{dU}{dt} = -I^2R$$

- The total energy is $U = \frac{Q^2}{2C} + \frac{1}{2}LI^2$

$$\frac{d}{dt}\left(\frac{Q^2}{2C} + \frac{1}{2}LI^2\right) = \frac{Q}{C}\frac{dQ}{dt} + LI\frac{dI}{dt}$$

- Set equations equal:

$$\frac{Q}{C}\frac{dQ}{dt} + LI\frac{dI}{dt} = -I^2R$$

- We know $\frac{dQ}{dt} = I$ and $\frac{dI}{dt} = \frac{d^2Q}{dt^2}$, Substitute these.

$$\frac{Q}{C}(I) + L(I)\frac{d^2Q}{dt^2} = -I^2R$$

- Divide the entire eqn by I (assuming $I \neq 0$).

$$\frac{Q}{C} + L\frac{d^2Q}{dt^2} = -IR, \text{ put } I = \frac{dQ}{dt}$$

$$\boxed{L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{1}{C}Q = 0}$$

This is the differential equation for an RLC circuit.

32.6 - The RLC Circuit

- **RLC Circuit**: A circuit containing a resistor (R), an inductor (L) and a capacitor (C) in series. Unlike the ideal LC circuit, the RLC circuit has a resistor. This resistor dissipates energy as heat ($P = I^2R$) from the circuit.

- **Damped Oscillation**: Because energy is being lost continuously, the oscillations of charge and current will not continue forever. The amplitude of the oscillations will exponentially decrease over time, eventually stopping. This is called damped oscillation.

- **Mechanical Analogy**:

  - The LC circuit was like a frictionless block-spring oscillator.

  - The RLC circuit is analogous to a block-spring oscillator with a damping force (like friction or air drag). The resistance R acts like the damping coefficient b.

- **Derivation: Differential Equation for an RLC Circuit**:-

• Analogy to a Damped Mechanical Oscillator :-

- This equation is mathematically identical to the equation for a damped harmonic oscillator in mechanics.

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$$

By comparing the two equations, we can eastablish the following analogies.

- Charge Q is analogous to position x.
- Inductance L is analogous to mass m.
- Resistance R is analogous to the damping coefficient b.
- Capacitance C is analogous to the inverse of the spring constant 1/k.

• Solution & Behaviour of the Circuit :-

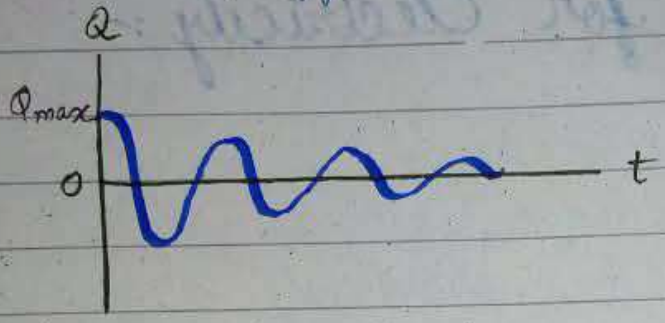
When the resistance R is small (specifically, when $R < \sqrt{4L/C}$), the system is underdamped and oscillates, but the amplitude of the oscillations decreases over time. The charge on the capacitor is given by :

$f = 2\pi w$

$\boxed{w = 2\pi f}$

$$Q(t) = Q_{max}\, e^{\frac{-Rt}{2L}} \cos(w_d t)$$

The term $e^{\frac{-Rt}{2L}}$ acts as an exponential "envelope" that causes the amplitude to decay, as seen in fig 32.16.



The damped angular frequency, w_d, is slightly lower than the natural frequency of an ideal LC circuit and is given by:

$$w_d = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$$

- Critically Damped: If the resistance has the value $R_c = \sqrt{4L/C}$ the circuit does not oscillate. The charge returns to zero in the quickest possible time without oscillating.

- Overdamped: When $R > R_c$, the system is overdamped and the charge slowly returns to zero without any oscillation.

34.2 - Maxwell's Equations

. These four equations are the absolute foundation of all classical phenomena involving electricity and magnetism. They unify these two forces into a single theory of electromagnetism.

1. Gauss's Law for Electricity :-

States that the total electric flux through a closed surface is equal to the charge enclosed inside the surface divided by the permittivity of free space.

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enclosed}}{\varepsilon_0}$$

2. Gauss's Law For Magnetism :-

States that the net magnetic flux through a closed loop is always zero, because magnetic field lines always form closed loops and magnetic monopoles do not exist.

$$\oint \vec{B} \cdot d\vec{A} = 0$$

3. Faraday's Law of Induction :-

A changing magnetic field creates an electric field. The line integral of the electric field around a closed loop (Emf) is equal to the rate of change of magnetic flux through the loop.

$$\oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$$

4. Ampere-Maxwell Law :-

A magnetic field is created by electric current and changing electric field (also known as displacement current).

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I + \epsilon_0 \mu_0 \frac{d\Phi_E}{dt}$$

. Lorentz Force Law :-

. This law is used in conjuction with Maxwell's equations. It decides the total force experienced by a charge q moving in the presence of both electric and magnetic fields

$$F = qE + q(v \times B)$$

Resolving components:

$$dE_y = dE \cos\theta$$

By using the pythagorean theorem, $\cos\theta = \frac{y}{r}$

$$= \frac{y}{(R^2+y^2)^{1/2}}.$$

Substituting the values dE and $\cos\theta$.

$$dE_y = \left[\frac{\lambda\, ds}{4\pi\epsilon_0(R^2+y^2)}\right] \times \left[\frac{y}{(R^2+y^2)^{1/2}}\right]$$

$$\therefore dE_y = \frac{\lambda y\, ds}{4\pi\epsilon_0(R^2+y^2)^{3/2}}$$

Applying integral:
To get the total field E_y, we integrate both sides.

$$\int dE_y = \int \frac{\lambda y}{4\pi\epsilon_0(R^2+y^2)^{3/2}}\, ds$$

Since λ, y, R and ϵ_0 are constant, we take them out of the integral:

$$E_y = \frac{\lambda y}{4\pi\epsilon_0(R^2+y^2)^{3/2}} \int ds$$

The integral $\int ds$ represents the total circumference on the ring, which is $2\pi R$.

Electric Field due to a Uniformly Charged Ring :-

Consider a ring of Radius 'R' carrying a uniform linear charge density 'λ' around its circumference. Consider a differential element of the ring length 'ds' located at an arbitary position at the ring. It contains an element of charge 'dq'.

The differential field 'dE' at point 'P' is given by Coulumb's law :

$$dE = \frac{dq}{4\pi\epsilon_0 r^2} \longrightarrow (1)$$

We know that linear charge density $\lambda = \frac{dq}{ds}$, therefore $dq = \lambda ds$.
From the geometry of the figure, the distance is $r^2 = R^2 + y^2$.

Substituting these values into eqⁿ (1).

$$dE = \frac{\lambda\, ds}{4\pi\epsilon_0 (R^2 + y^2)}$$

To find the resultant field at point 'P', we must add up all vector field contributions 'dE' made by the differential element of the ring. There is only one component to the total field which is along the y-axis, while components along the x-axis and z-axis are equal to zero due to symmetry.

Energy Stored in a Capacitor :-

Consider a capacitor of capacitance 'C' being charged. At any intermediate instant during the charging process, let 'q' be the charge on the capacitor and 'V' be the potential difference between the plates.

From the definition of capacitance:

$$V = \frac{q}{C}$$

The work done 'dW' in transferring an additional small amount of charge 'dq' from the negative plate to the positive plate against the potential difference 'V' is given the definition of electric potential:

$$V = \frac{dW}{dq}$$

$$dW = V \cdot dq \qquad \because V = q/C$$

$$dW = \left(\frac{q}{C}\right) dq$$

The total work done 'W' in charging the capacitor from an initial charge 0 to a final charge 'Q' is obtained by integrating the equation:

$$W = \int_0^Q dW = \int_0^Q \frac{q}{C}\, dq$$

$$W = \frac{1}{C} \int_0^Q q \cdot dq$$

Since the total charge $q = \lambda(2\pi R)$:

$$\boxed{E_y = \frac{qy}{4\pi\epsilon_0(R^2+y^2)^{3/2}}}$$

Special Case: If the point P is very far from the center ($y >> R$), then R^2 can be neglected in comparison to y^2.

$$E_y = \frac{qy}{4\pi\epsilon_0(0+y^2)^{3/2}} = \frac{qy}{4\pi\epsilon_0(y^2)^{3/2}}$$

$$E_y = \frac{q}{4\pi\epsilon_0 y^2}$$

This confirms that at large distance, the ring behaves like a point charge.

Electric Field Due to a Charged Disk :-

Consider a circular plastic disk of radius R with a uniform positive surface charge density σ. We want to find the electric field at a point P on the central axis at a distance z from the center.

We treat the disk as a collection of concentric thin rings. Consider one such ring with radius r and radial with dr.
The electric field dE due to this ring at distance z is given by (Ring Formula).

$$dE = \frac{z \cdot dq}{4\pi \epsilon_0 (z^2 + r^2)^{3/2}} \longrightarrow (1)$$

The area of the thin ring is its circumference multiplied by its width:

$$dA = (2\pi r)\, dr$$

Since σ = charge/area, the charge dq is:

$$dq = \sigma\, dA = \sigma (2\pi r\, dr)$$

Substitute the value of dq into eq^n (1) and integrate from $r = 0$ to $r = R$.

$$E = \int dE = \int_0^R \frac{z\,(\sigma\, 2\pi r\, dr)}{4\pi \epsilon_0 (z^2 + r^2)^{3/2}}$$

$$W = \frac{1}{C}\left[\frac{q^2}{2}\right]_0^Q$$

$$W = \frac{1}{C}\left(\frac{Q^2}{2} - 0\right)$$

$$W = \frac{Q^2}{2C}$$

This work done is stored in the capacitor in the form of electrostatic potential energy 'U'.

$$\boxed{U = \frac{Q^2}{2C}}$$

We can also express this energy in terms of Potential Difference (V). Using the relation $Q = CV$,

$$U = \frac{C^2 V^2}{2C}$$

$$\boxed{U = \frac{1}{2} CV^2}$$

We define Energy Density (u) as Energy per unit Volume.

$$u = \frac{U}{\text{Volume}}$$

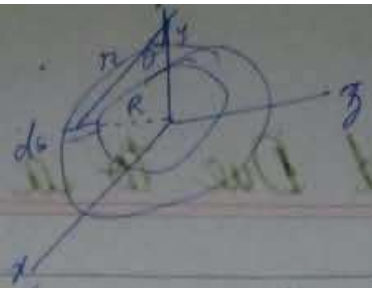
$$u = \frac{\frac{1}{2}\left(\frac{\epsilon_0 A}{d}\right)(Ed)^2}{A \cdot d}$$

$$\boxed{u = \frac{1}{2}\epsilon_0 E^2}$$

$\because$ Volume $= A \cdot d$

$\because C = \frac{\epsilon_0 A}{d}$

$\because V = Ed$



Take the constants $(\sigma, z, \pi, \epsilon_0)$ out of the $\int$

$$E = \frac{\sigma z}{4\epsilon_0} \int_0^R (z^2 + r^2)^{-3/2} (2r)\, dr.$$

$$E = \frac{\sigma z}{4\epsilon_0} \left[\frac{(z^2+r^2)^{-1/2}}{-1/2} \right]_0^R$$

$$E = \frac{\sigma z (-2)}{4\epsilon_0} \left[\frac{1}{\sqrt{z^2+r^2}} \right]_0^R$$

$$E = \frac{-\sigma z}{2\epsilon_0} \left[\frac{1}{\sqrt{z^2+R^2}} - \frac{1}{\sqrt{z^2+0}} \right]$$

$$E = \frac{\sigma z}{2\epsilon_0} \left[\frac{1}{z} - \frac{1}{\sqrt{z^2+R^2}} \right]$$

$$\boxed{E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2+R^2}} \right)}$$

Special Case: If the radius of the disk becomes very large ($R \gg z$) the term $\frac{z}{\sqrt{z^2+R^2}}$ approches 0.

$$E = \frac{\sigma}{2\epsilon_0} (1-0)$$

$$\boxed{E = \frac{\sigma}{2\epsilon_0}}$$